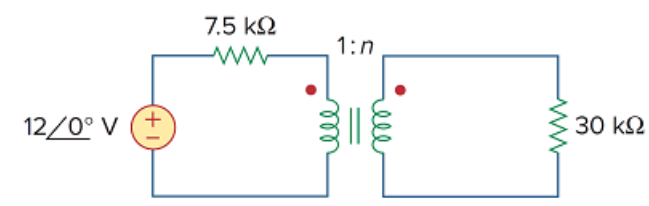


## Problem

For the circuit in Fig. 13.129, find the turns ratio so that the maximum power is delivered to the 30-k $\Omega$  resistor.

Figure 13.129



## Step-by-step solution

## Step 1 of 2

Refer to Figure 13.129 in the textbook for a transformer network.

Reflect the secondary impedance of transformer to the primary side.

$$R_L = \frac{R_{L2}}{n^2}$$

Here,  $n$  is the turns ratio of the transformer. Therefore,

$$\begin{aligned} R_L &= \frac{30 \text{ k}\Omega}{n^2} \\ &= \frac{30000}{n^2} \end{aligned}$$

Draw the simplified circuit.

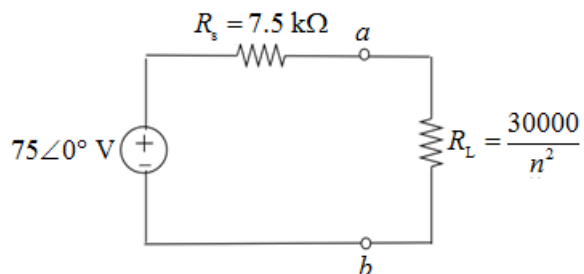


Figure 1

## Step 2 of 2

Consider that the maximum power is transferred to the load when the load resistance is equal to source resistance as per the maximum power transfer theorem. Therefore,

$$R_L = R_s$$

Substitute the values of  $R_L$  and  $R_s$ .

$$\frac{30 \text{ k}\Omega}{n^2} = 7.5 \text{ k}\Omega$$

$$\begin{aligned} n^2 &= \frac{30 \text{ k}\Omega}{7.5 \text{ k}\Omega} \\ &= 4 \end{aligned}$$

$$n = 2$$

Therefore, the value of turns ratio,  $n$  for maximum power transfer in the circuit is 2.